S1 Supplementary Methods

New data products, the cross-carrier phase-locking helper, the two new onset-hazard null constructions, and the intrinsic-strength interaction model

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S1.0 Dataset and relationship to the companion articles

This article analyzes RichSleep ((RichSleep contributors, 2026), 24 subjects, whole-night PSG) exclusively - the identical dataset and sleep-spindle catalog used in both companion articles' (Langell, 2026a; 2026b) analyses. NMD's 2026-07-12/13 delivery adds six new data products, validated end-to-end in `01_validate_v3_delivery.py` (`S1.1`). A bit-identity check on all 13 columns shared between `phase_resolved_v3` and the companion articles' `event_phase_v2` confirms geometry, onset/peak times, and phase values are byte-identical between deliveries - every geometry number reused from the companion articles in this article is therefore the companion articles' own frozen number, not a recomputation.

S1.1 New and reused data products

Product	Description
`phase_resolved_v3`	24 subjects, 23,029 N2-spindle events (identical catalog to `event_phase_v2`), now 33 columns: adds `frequency_hz`, `amplitude`, `sigma_power`, `sigma_power_z_n2` (intrinsic strength) and `so_amplitude`, `so_spindle_coupling_score`, `so_spindle_latency_sec`, `so_partner_missing`, `so_coupling_qc_flag` (SO-spindle coupling).
`slow_oscillation_v1`	68,386 raw SO events (26,377 N2 + 42,009 N3), no phase columns - the source catalog for both the v3 SO-coupling columns and the N3-SO phase-resolved export below.
`phase_continuous_v1`	24 subjects, 8,834,120 rows total (10.2 h/subject), continuous 10 Hz `phi_cardiac`/`phi_resp`/`phi_rem_theta`; `cardiac_phase_valid`/`resp_phase_valid`≈100%, `rem_theta_valid`≈8.2% (REM only). The genuine background/risk-set stream used by Test B (`S1.3`).
`event_phase_n3_so_v1`	42,009 N3-SO events, phase-resolved against cardiac and respiration at both SO trough and SO upstate (100% valid/QC-ok).
`event_phase_rem_theta_v1`	2,412 REM-theta 30 s epochs, phase-resolved against cardiac, respiration, and theta (100% valid; uneven per-subject N, 2-219).
`non_event_risk_v1`	66,827 N2-restricted, per-subject-matched, margin-excluded (±30 s around every real spindle onset) non-event timepoints, each carrying its own `phi_cardiac`/`phi_resp` - NMD's purpose-built risk set for case-control hazard-style testing (Test A, `S1.3`).

Table 1: All six data products, `01_validate_v3_delivery.py`. Geometry (`coords_9d`) exists only for the companion articles' N2-spindle-locked event/control/pre_far bins - see `S3.0` for the resulting sample-coverage implications.

Only 2,708/23,029 spindles (11.8%) have a matched SO partner (`so_partner_missing`= 0); of the companion articles' 190 geometry-valid events, only 25 (across 15 subjects) are SO-coupled - the power caveat behind R5′s descriptive-only framing (`S1.5`, part 3).

S1.2 Harmonic statistic and subject-rotation permutation null (unchanged from the companion articles)

For a set of phases $\varphi_1, ..., \varphi_n$ (radians), the first- and second-harmonic mean resultant lengths and the omnibus statistic used throughout the cross-carrier sweep and Test A/B:

$$R_1 = |\langle e^{i\varphi} \rangle| = \sqrt{\left(\frac{1}{n}\sum_k \cos\varphi_k\right)^2 + \left(\frac{1}{n}\sum_k \sin\varphi_k\right)^2} \tag{1}$$

$$R_2 = |\langle e^{2i\varphi} \rangle|, \quad T_2 = R_1^2 + R_2^2 \tag{2}$$

T_2 is an honestly-labelled analogue of Hermans-Rasson's robustness-to-multimodality motivation, not a re-implementation of that specific literature statistic (unchanged rationale from the companion article's `S1.2`) - every p-value here is a Monte Carlo permutation p-value, not an asymptotic one.

Subject-rotation permutation test (`rvc_sleep_support.subject_rotation_permutation_test`, unchanged from the companion article): for $n_{\text{perm}} = 2000$ permutations, draw one i.i.d. $\text{Uniform}(0, 2\pi)$ rotation per subject and add it to **all** of that subject's phases; recompute R_1, T_2 on the rotated pooled sample each time. This destroys any common population-level preferred phase while exactly preserving each subject's event count and within-subject clustering. Reported p-values: $p_{R_1} = \left(1 + \#\left\{k : R_1^{(k)} \geq R_1^{\text{obs}}\right\}\right)/(1 + n_{\text{perm}})$, and identically for T_2.

Per-subject and second-order tests (`rvc_sleep_support.run_phase_locking_family`, the generic helper shared across all five carrier/reference-point conditions in `S1.3`): a per-subject Rayleigh test (Zar 1999 large-sample approximation, valid for $n \approx 10$ or more) for every subject with ≥ 1 event, plus a second-order Rayleigh test on the subset of subjects with ≥ 5 events' circular mean directions (`min_events_per_subject=5`) - the per-subject table therefore always has N=24 (23 for REM-theta, one subject with < 5 epochs), while the second-order test's N can be smaller when a carrier has thinly-sampled subjects.

S1.3 Cross-carrier phase-locking sweep and the two new onset-hazard tests

Cross-carrier sweep (`03_multistage_phase_locking.py`): `run_phase_locking_family` (S1.2) is applied to five carrier/reference-point conditions - N2-spindle onset (`phi_cardiac_onset`/`phi_resp_onset`), N2-spindle peak (`phi_cardiac_peak`/`phi_resp_peak`), N3-SO trough (`phi_cardiac_trough`/`phi_resp_trough`), N3-SO upstate (`phi_cardiac_upstate`/`phi_resp_upstate`), and REM-theta (`phi_cardiac`/`phi_resp`) - each at both channels, 10 tests total. The primary family for BH-FDR ((Benjamini & Hochberg, 1995), $\alpha = 0.05$) is these 10 pooled subject-rotation-permutation tests on T_2 (and, separately, on R_1); the second-order Rayleigh p-values are BH-FDR corrected separately within their own 10-test family. REM-theta's per-subject/second-order arm excludes the one subject with < 5 epochs; the pooled/permutation arm retains all 2,412 events regardless.

Test A - case-control logistic regression (`04_spindle_onset_hazard_model.py`, `rvc_sleep_support.fit_logit_cluster`): real N2-spindle onsets ($n = 23029$, `is_case`$= 1$) are pooled with `non_event_risk_v1` rows ($n = 66827$, `is_case`$= 0$); `is_case ~ cos(phase) + sin(phase)`, logistic regression with cluster-robust SE by subject, joint Wald $\chi^2(2)$ test on the $(\cos, \sin)$ coefficient pair against the null that both are zero. The implied preferred phase is $\hat{\varphi} = \text{atan2}(b_{\sin}, b_{\cos})$. A significant result means real onsets occur at different phases than a matched, N2-restricted, subject-balanced sample of "could a spindle have started here" timepoints - a genuine hazard-ratio-style test using the real (non-uniform) N2 exposure distribution, not an assumed-uniform one.

Test B - circular time-shift surrogate (`rvc_sleep_support.circular_time_shift_surrogate_test`): for $n_{\text{perm}} = 500$ permutations, each subject's own continuous phase stream (`phase_continuous_v1`) is shifted by an independent random lag (wrapping circularly at that subject's own recording duration, minimum shift 60 s) and phase is re-read at that subject's real onset clock-times under the shifted stream; T_2 (and R_1) are recomputed each time to build the null distribution. This preserves the exact non-sinusoidal waveform and autocorrelation structure of each subject's own physiological signal - a different null-construction principle from both Test A and the subject-rotation permutation (Null 0 in the main text and `S1.2`), sharing no free parameters or resampling unit with either.

An initial attempt at a third exposure model - a phase-binned Poisson-offset GLM using the per-subject geometry-run H5′s `labels/stage` array as a continuous hypnogram - was abandoned when that array turned out to be populated only at 177 "jacobian-center" windows per subject (a sparse local-dynamics-fit subset), undercounting true N2 exposure by roughly two orders of magnitude; documented in

handover/handover_From_nmd_richsleep_eap_v2_delivery_received.md S6. non_event_risk_v1 (Test A) supersedes this abandoned approach.

S1.4 Held-out population-preferred peak phase (self-contained recomputation)

Recomputed here, from this article's own phase_resolved_v3 catalog, for self-containment (05_qe_interaction_moderation.py): using the (subject_id, onset_sec) key, anti-join the companion articles' 190 geometry-valid events out of the full 23,029-event catalog, leaving 22,839 held-out events. For each of the 24 subjects, compute their own circular mean direction of phi_resp_peak on the held-out set; $\mu_{\text{resp,peak}}$ is the second-order circular mean (Equation 2′s R_1 construction applied to the 24 subject-mean angles) of those 24 held-out subject means. This reproduces the companion article's own held-out estimate exactly (both use the identical anti-join logic on bit-identical data).

S1.5 Intrinsic-strength (qe) × alignment interaction model (R4-R5)

1. **Alignment score.** $\text{resp_peak_alignment}_e = \cos\left(\varphi_{\text{resp,peak},e} - \mu_{\text{resp,peak}}\right)$ for each of the companion articles' 190 geometry-valid events e (S1.4).
2. **qe operationalizations.** Primary: $\text{qe} = \log(1 + \text{sigma_power_z_n2})$ (raw sigma_power_z_n2 is heavily right-skewed, skew ≈ 2.75). Robustness: $\text{qe} = \log(1 + \text{amplitude})$ and untransformed sigma_power_z_n2.
3. **Primary/robustness family.** delta ~ resp_peak_alignment * qe + covariates via fit_wls_cluster (weights $1/n_{\text{events,subject}}$, cluster-robust SE by subject; all terms mean-centered before forming the product term, so the reported _interaction coefficient is a genuine interaction, not confounded with either main effect's scale). delta is event_bin_value - matched_control_median (event_control mode) or event_bin_value - mean(pre_far bins, same event) (event_prefar mode), for each of the 9 primary geometry endpoints (m, d, e, d_n, d_l, d_s, e_e, e_s, e_m, inherited unchanged from the companion articles' coords_9d). 9 endpoints × 2 delta modes × 3 qe operationalizations - the **interaction** coefficient's BH-FDR correction is applied within each of the three 18-test families separately (one per qe operationalization), matching the companion article's Mål C family-size convention. Nuisance covariates: event_duration_sec, window_center_sec, ocular_stability_index, sympathetic_index, vagal_index, eog_quality_quality, ecg_quality_quality, resp_quality_quality, cardioresp_quality_quality, anchor_support_fraction_quality (available_covariates() requires $\geq 50\%$ coverage).
4. **Secondary/exploratory SO-coupled check.** On the subset with so_partner_missing= 0 (25 events/15 subjects before delta-mode-specific attrition; 21/14 in event_control mode, 19/13 in event_prefar mode, matching the same per-mode covariate-availability pattern as the primary family's 172/160), Pearson correlations of delta against so_spindle_coupling_score and so_amplitude, for each of the 9 endpoints × 2 delta modes = 18 pairs. No interaction model, no FDR family - explicitly too few degrees of freedom for a confirmatory test (S1.5 part 3′s power caveat).

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